

Java OOP Banking Case Study (Without Collections)

Scenario

ABC Bank is developing a Java-based Banking Management System for new employees to learn Object-Oriented Programming. The application should manage one customer and one account at a time.

Do not use collections such as ArrayList, HashMap, Set, or Map.

Business Requirements

- Register a customer
- Open Savings, Current, or Fixed Deposit account
- Deposit money
- Withdraw money
- Display account details
- Calculate interest
- Validate account operations

Sample Input

Customer ID: C101
Customer Name: Rahul Sharma
Account Number: 100101
Account Type: Savings
Opening Balance: 50000
Deposit Amount: 5000
Withdrawal Amount: 3000

Programming Tasks (Questions Only)

1. Identify the classes required for the application.
2. Identify suitable attributes and methods for each class.
3. Create an Account superclass.
4. Create SavingsAccount, CurrentAccount, and FixedDepositAccount classes using inheritance.
5. Implement encapsulation using private data members.
6. Create constructors to initialize account details.
7. Override the withdraw() method for each account type.
8. Overload the deposit() method using different parameter lists.
9. Create an abstract Account class with calculateInterest().
10. Create a BankOperations interface with deposit(), withdraw(), and checkBalance().
11. Demonstrate runtime polymorphism using an Account reference.
12. Validate deposits and withdrawals using exception handling.
13. Prevent Savings Account balance from going below the minimum balance.
14. Prevent premature withdrawal from a Fixed Deposit account.
15. Display account details using an overridden toString() method.
16. Search for an account using the account number (single object comparison).
17. Calculate yearly interest for the account.
18. Generate a simple transaction receipt.
19. Draw the UML class diagram.
20. Explain where encapsulation, inheritance, abstraction, and polymorphism are used in your solution.

Expected Output

Customer Registered Successfully
Account Created Successfully
Deposit Successful
Updated Balance: 52000
Withdrawal Successful
Updated Balance: 49000
Interest Calculated Successfully
Account Details Displayed Successfully

Java Concepts Covered

Classes & Objects
Constructors
Encapsulation
Inheritance
Method Overloading
Method Overriding
Runtime Polymorphism
Abstraction
Interfaces
Exception Handling
toString()
final Keyword
Static Members